

PREPARED FOR

T6N, R4W, Mount Diablo Meridian

COORDINATES 38.3459°N 122.3047°W WGS84	ELEVATION 66 ft NAVD88 · MSL	STATE PLANE CA 2 (0402) · N 1,887,922.81 E 6,474,289.05 (US ft, NAD83) -	FEMA ZONE Zone X 06055C0510FF	DECLINATION 13.0° E WMM 2026
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APN 038050009000 | ZONING AP | ACRES 0.481 ac

BEST GNSS WINDOW Sun, Jul 12 · 2:00 PM – 6:00 PM | PDOP 0.97 · 35 sv | NEAREST CORS
P261 · 14.1 mi ✓

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Site file · Google Earth
Download KMZ

RECORDER
Napa County Recorder-Clerk · 1127 First
St, Napa, CA 94559

NEAREST CORS
P261 · 14.1 mi

GENERATED
July 12, 2026

ASSESSOR
Napa County Assessor · 1127 First St,
Napa, CA 94559

NEAREST BM
JT9010 · 0.2 mi

SYSTEM
TopoScout · info@toposcout.com

This report compiles data from public and third-party sources (NGS, FEMA, USGS, NOAA, and county records) current as of the generation date shown above. GNSS planning values are predictive and conditions may change. This document is provided for planning purposes only and does not constitute a boundary survey, legal land description, or engineering certification. Verify all critical values against primary sources.

SITE & SIGNAL REPORT

COORDINATES

38.3459, -122.3047

WGS84 / NAD83

ELEVATION

66 ft

Above MSL (approx)

MAGNETIC DECLINATION

13.0° E

Compass reads 13.0° too far W

FEMA FLOOD ZONE

X

AREA OF MINIMAL FLOOD HAZARD · Panel 06055C0510FF

IONOSPHERIC CONDITIONS

PEAK KP INDEX

4.7

AVG KP

2.6

SOLAR FLUX F10.7

138 sfu

TEC (VTEC) ▲ DEGRADED

20.6 TECU

-4.8 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.6). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Sun, Jul 12	0.97	2:00 PM – 6:00 PM	35	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates. For site-specific weather, terrain mask angle, and nearest CORS station, order an On-Demand Site Report.



WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 12, 2026) · Conditions may change – verify before fieldwork

✖ GPS CONSTELLATION HEALTH

▲ 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.9–0.9 across survey window
PDOP remains ≤ 0.9 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

Sunday, Jul 12

WEEKEND

Overcast

10 mph

GO

High 90°F Low 56°F Wind max 10 mph Gusts 12 mph Precip 0.00" Sunrise 5:56 AM Sunset 8:33 PM

Waning Crescent 6%

GNSS: Avg PDOP 0.97 · Best 4-hr window 2:00 PM – 6:00 PM (PDOP 0.93, 35 sats avg)

GNSS / DOP - HOURLY					
	PDOP		SATS		RATING
6:00 AM	0.87		36 sv		IDEAL
7:00 AM	0.91		34 sv		IDEAL
8:00 AM	1.06		29 sv		GOOD
9:00 AM	1.00		33 sv		GOOD
10:00 AM	1.00		29 sv		GOOD
11:00 AM	1.06		27 sv		GOOD
12:00 PM	0.96		29 sv		IDEAL
1:00 PM	1.00		28 sv		GOOD
2:00 PM	0.90		34 sv		IDEAL
3:00 PM	0.98		33 sv		IDEAL
4:00 PM	0.98		34 sv		IDEAL
5:00 PM	0.86		38 sv		IDEAL
6:00 PM	1.01		32 sv		GOOD
7:00 PM	1.04		30 sv		GOOD



GO — GOOD CONDITIONS

Partly cloudy | Wind: 9.5 mph (gusts 11.6 mph) | Precip: 0" | Temp: 56.3°–89.7°F



DRONE WIND ANALYSIS



GO Peak: 9.5/11.6 mph SSE

* Sunrise: 5:56 AM Sunset: 8:33 PM fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤=turbulence	DIR
5 AM	3/5 mph		SSE
6 AM	4/7 mph		ESE
7 AM	2/3 mph		ESE
8 AM	3/3 mph		W
9 AM	3/4 mph		SW
10 AM	5/5 mph		SSW
11 AM	6/7 mph		SSE
12 PM	5/6 mph		S
1 PM	4/5 mph		SE
2 PM	6/7 mph		S
3 PM	7/8 mph		S
4 PM	8/10 mph		S
5 PM	10/11 mph		S
6 PM	9/11 mph		SSW
7 PM	9/12 mph		S
8 PM	6/11 mph		SSE

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ► FREE TO FLY
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#) ↗
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

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FIELD PLANNING REPORT — GNSS SURVEY OPERATIONS

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JOB: N/A

SITE: Napa County, California (38.3459, -122.3047)

REPORT DATE: Sunday, July 12, 2026

SURVEY START: Sunday, July 12, 2026

ANALYSIS WINDOW: 1 day

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VERDICT

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GO — Proceed with survey operations.

Primary reason: Excellent PDOP geometry (avg 0.97, best window 0.93) combined with favorable weather and moderate ionospheric conditions support high-confidence static and RTK operations.

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BEST SURVEY WINDOW

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2:00 PM – 6:00 PM LOCAL TIME (Sunday, July 12, 2026)

Why: PDOP reaches optimal 0.93 during this window with 35 satellites in view (multi-constellation). Excellent geometry for both static baselines and real-time kinematic work. Weather is overcast but dry (0.00" precip) with manageable 10 mph max wind.

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BEST 4-HOUR STATIC GNSS WINDOW

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2:00 PM – 6:00 PM LOCAL TIME

Recommended occupation: 120 minutes minimum per setup (ionospheric TEC is 4.8 TECU below 30-day median; moderate geomagnetic activity Kp 4.7 suggests conservative approach). Four-hour window permits two independent static sessions or one extended occupation with redundancy. Use 15° elevation mask; PDOP 0.93 sustained throughout.

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DRONE OPERATIONS ASSESSMENT

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FAVORABLE. Visual line-of-sight conditions expected despite overcast sky. Wind max 10 mph is within safe operating limits for most UAS platforms. No precipitation risk. Ensure airspace coordination; Napa County is near controlled airspace (Oakland Class C, approximately 60 nm south). Brief all drone pilots on GPS/GNSS receiver performance degradation in moderate ionospheric activity — expect slightly larger RTK initialization times on baseline starts.

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FIELD SAFETY WARNINGS

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MINOR ALERT — Ionospheric TEC quality flagged (non-zero quality_flag). Treat TEC estimate (-4.8 TECU anomaly) as reference only; absolute values less reliable. No impact on field operations but note for post-processing QA.

SITE CONTEXT ADVISORIES (non-ranking factors):

- GPS PRN 11 currently unhealthy (UNUSUFN). Receivers will auto-exclude. Constellation remains robust (35+ sats in best window).

- Wildfire status: All-clear within reporting range.
- Land cover (Dwarf Scrub) and recommended terrain elevation mask (13°) noted for multipath mitigation; sky obstruction risk is low.
- Air quality: AQI 45 (Good). Field crew conditions acceptable.

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SUMMARY

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Sunday, July 12, 2026 is operationally GO for GNSS survey work in Napa County. The 2:00 PM – 6:00 PM window offers superior satellite geometry (PDOP 0.93, 35 sats) with stable weather (overcast, dry, 10 mph winds) and moderate ionospheric conditions (Kp 4.7, F10.7 138 sfu). Plan static occupations at 120 minutes minimum to account for ionospheric delay; RTK and real-time applications are feasible with standard initialization protocols. Drone operations are favorable. Monitor GPS PRN 11 health status for future campaign planning; present constellation provides redundancy. Site conditions (scrub terrain, low multipath) support accuracy targets. Proceed with survey.

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END OF REPORT

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COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Napa County Recorder-Clerk
1127 First St, Napa, CA 94559

ASSESSOR / TAX OFFICE

Napa County Assessor
1127 First St, Napa, CA 94559

CORS STATIONS – WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
NGS ↗	P261	P261	14.1 mi
NGS ↗	P198	P198	17.5 mi
NGS ↗	OHLN	OHLN	23.5 mi
NGS ↗	P196	P196	24 mi
NGS ↗	CASR	CASR	24.9 mi
NGS ↗	P267	P267	26.3 mi

NGS BENCHMARKS – WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
 Datasheet ↗	JT9010	"13 5"	19.10m / 62.7ft NAVD88	undefined	0.2 mi
 Datasheet ↗	JT9018	"13 E"	17.97m / 59.0ft NAVD88	undefined	0.2 mi
 Datasheet ↗	JT9009	"13 4"	17.90m / 58.7ft NAVD88	undefined	0.2 mi
 Datasheet ↗	JT9017	"13 D 1"	17.89m / 58.7ft NAVD88	undefined	0.2 mi
 Datasheet ↗	JT9034	"20 A"	20.73m / 68.0ft NAVD88	undefined	0.3 mi
 Datasheet ↗	JT9012	"13 6"	19.06m / 62.5ft NAVD88	undefined	0.4 mi
 Datasheet ↗	JT9043	"25 A"	18.71m / 61.4ft NAVD88	undefined	0.4 mi
 Datasheet ↗	JT9016	"13 D"	18.20m / 59.7ft NAVD88	undefined	0.5 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet					

FEMA FLOOD ZONE

X – AREA OF MINIMAL FLOOD HAZARD

Panel: 06055C0510FF

Effective: Sep 29, 2010

[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

Copy All

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector

Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

PID	NAME	LAT	LON	ELEV_FT	ORDER
JT9010	13 5	38.348333	-122.302778	62.7	
JT9018	13 E	38.348333	-122.302778	59.0	
JT9009	13 4	38.346667	-122.301111	58.7	
JT9017	13 D 1	38.345000	-122.301111	58.7	
JT9034	20 A	38.345000	-122.309444	68.0	
JT9012	13 6	38.351667	-122.304444	62.5	
JT9043	25 A	38.340000	-122.302778	61.4	
JT9016	13 D	38.341667	-122.297778	59.7	

USGS STREAM GAUGES – within 25 miles

STATION	DIST	LEVEL	FLOW	FLOOD STATUS
NAPA R NR NAPA CA #11458000	1.5 mi	2.36 ft	4.08 cfs	● Normal
NATHANSON C A SONOMA CA #11458600	9.3 mi	—	—	● Normal
SONOMA C A AGUA CALIENTE CA #11458500	10.4 mi	8.84 ft	2.73 cfs	● Normal
NAPA R NR ST HELENA CA #11456000	14.1 mi	1.44 ft	3.43 cfs	● Normal
SONOMA CREEK A KENWOOD CA #11458433	14.9 mi	3.93 ft	0.19 cfs	● Normal
PUTAH C NR WINTERS CA #11454000	16.8 mi	7.85 ft	578 cfs	— Unknown
SUISUN SLOUGH NR BENICIA CA #11455740	19.6 mi	14.69 ft	-3,170 cfs	— Unknown
POPE C A WALTER SPRINGS CA #11453590	21.1 mi	—	—	— Unknown

USGS LAKES & RESERVOIRS – within 150 miles

STATION	DIST	LEVEL	FULL POOL	LINK
CLEAR LK A LAKEPORT CA #11450000	58.1 mi	5.07 ft	N/R	USGS ↗

CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Del Valle	60.6 mi	697.01 ft (-9 ft)	706 ft	36K af	68%
Folsom Lake	66.5 mi	448.64 ft (-17 ft)	466 ft	792K af	81%
Lake Oroville	93.4 mi	862.11 ft (-38 ft)	900 ft	2,871K af	81%
New Melones Lake	100.1 mi	1,020.25 ft (-68 ft)	1,088 ft	1,664K af	69%
Black Butte Lake	101.2 mi	—	228 ft	74K af	55%

USBR RESERVOIRS (Bureau of Reclamation) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
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ARMY CORPS RESERVOIRS (USACE) – within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Del Valle	64 mi	697 ft	695.01 ft	—	100%
Camanche	89 mi	230.3 ft	221.86 ft	—	104%
Englebright Lake-Pool	95 mi	525.1 ft	522.35 ft	—	101%
Farmington Dam-Pool	99 mi	124.9 ft	117.09 ft	—	107%
Black Butte-Pool	101 mi	457 ft	458.36 ft	—	100%
Don Pedro	137 mi	—	N/R	—	—

USGS GROUNDWATER WELLS – within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
005N004W32A002M	7 mi	200 ft	87.71 ft BGS	2014-11-05	N100CACSTL
005N004W28R001M	7 mi	51 ft	41.90 ft BGS	1977-10-03	N100CACSTL
004N004W04C001M	8 mi	80 ft	11.20 ft BGS	1977-10-03	N100CACSTL
004N004W05B001M	8 mi	—	54.10 ft BGS	1977-10-03	N100CACSTL
005N004W31R001M	8.1 mi	295 ft	—	—	—
004N004W05D002M	8.3 mi	60 ft	14.60 ft BGS	1977-10-03	N100CACSTL

004N004W06B001M	8.3 mi	201 ft	21.86 ft BGS	2022-06-07	—
004N004W02L001M	8.8 mi	—	9.90 ft BGS	1977-10-02	N100CACSTL

🕒 AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The Napa Valley Coastal Range aquifer (N100CACSTL) consists of Pliocene-Pleistocene alluvial fan and fluvial deposits laid down by ancestral drainages during alternating sea-level cycles. These unconsolidated to semi-consolidated sands, gravels, and silts provide moderate storage and transmissivity in a structurally complex basin. Water availability at this site depends on local recharge and position within the valley's groundwater flow system; the wide scatter in water-level measurements suggests proximity to discharge areas or influence from seasonal pumping.

FORMATION

Name	Napa Valley Coastal Range Quaternary Alluvium
Age / Rock	Pliocene–Pleistocene alluvial fan and fluvial sands, gravels, and silts
Extent	300–500 sq mi (est. USGS/DWR)

DEPTH & RECHARGE

Depth Range	51–295 ft (5–30 floors)
Typical Depth	100–150 ft (10–15 floors)
Recharge Source	Precipitation on upland ridges + lateral flow from Mayacamas Range + stream losing reaches
Transit Time	15–40 years to 100 ft depth

WATER QUALITY

pH	est. 7.0–7.6 (est. USGS/DWR)
Hardness	est. 180–350 mg/L CaCO3 (est. USGS/DWR)
TDS	est. 280–550 mg/L (est. USGS/DWR)
Notes	No sample data — typical for formation. Iron and manganese may be elevated in reducing zones; localized nitrate possible near agricultural areas.

TREND

Trend	Decline evident: water levels in 1977 ranged 9.90–54.10 ft BGS; 2014 and 2022 readings show 21.86–87.71 ft BGS, indicating net long-term drawdown of 20–40 ft at this location
Source	USGS NWIS historical record
Seasonal Swing	8–20 ft (estimated from sparse seasonal pairs; autumn 1977 vs. June 2022 span suggests multi-year trend dominates over seasonal signal)

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ LOW SUBSIDENCE RISK — No known critically overdrafted basin

Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

🕒 AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU

The North Coast Coastal Ranges aquifer system (N100CACSTL) in this region is characterized by well-drained soils and moderate groundwater utilization not exceeding sustainable yield, resulting in minimal subsidence risk. The area's geological setting features consolidated bedrock and alluvial deposits with adequate natural recharge, preventing the excessive groundwater withdrawal that typically triggers clay layer compaction. Surveyors working in this location can rely on historically stable bench marks with confidence that published elevations remain valid without subsidence corrections.

SUBSIDENCE ASSESSMENT

Susceptibility	Low
Primary Mechanism	None - consolidated terrain with balanced recharge-discharge
Annual Rate	<1 mm/yr (est. USGS)
Historic Loss	None documented
Aquifer Compaction	None - aquifer not over-pumped
SGMA Status	Not applicable - not in critically overdrafted basin

BENCHMARK IMPLICATIONS

Monument Risk	Low - monuments stable
Elevation Validity	Good - no active subsidence
GNSS Recommendation	Standard benchmark verification adequate

SURVEY ACTION

Required Action	No special action required
Dataset	CA DWR SGMA Bulletin 118 2024.1
Coverage	California critically overdrafted basins only - full US coverage pending

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)		WIND / SNOW / ICE / FROST (ASCE 7-22)	
Site Class	D (default)	Basic Wind Speed	92 mph
Risk Category	II	Ground Snow Load	5.8 psf
		Winter Wind (W2)	0.35
		Ice Thickness	None
		Frost Depth	12 in minimum
		Est. from IECC Zone 3C — AFI data unavailable Verify with local building official before construction	
		Source: ASCE 7-22 Hazard Tool · gis.asce.org	

AI CLIMATE SUMMARY — RESIDENTIAL DESIGN

This mild marine climate requires a moderate performance envelope—you'll want good air sealing and some insulation, but extreme R-values aren't critical given the temperate year-round conditions. Maximize south-facing glazing for passive solar gain in winter, but protect west-facing windows with overhangs or vegetation since summer sun can still overheat, and minimize east-facing glass to avoid morning heat gain. The single most effective passive strategy here is cross-ventilation: design operable windows to capture coastal breezes during the frequent mild months, which will naturally cool the home and reduce HVAC loads substantially.

AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU

This coastal marine site in IECC Zone 3C presents a well-drained gravelly loam foundation with moderate wind exposure and light snow loads. Seismic parameters are currently unavailable—a geotechnical investigation may warrant confirmation of site class and seismic design category before finalizing structural systems. Wind design at 92 mph and frost depth of approximately 12 inches should guide foundation depth and detailing.

FOUNDATION

Type Conventional spread footing anticipated, subject to geotechnical confirmation

Basis Well-drained Coombs gravelly loam suggests adequate bearing capacity for conventional foundations; however, seismic design category cannot be assessed without confirmed Ss and S1 values. Consult a licensed engineer and geotechnical professional to confirm site class and bearing capacity.

STRUCTURAL IMPLICATIONS

Seismic Seismic design category indeterminate—SDC values are null. A geotechnical investigation is strongly suggested to establish site class and obtain USGS seismic parameters. Structural framing detailing may warrant special consideration once SDC is confirmed.

Wind Exposure Exposure C (open terrain) confirmed per ASCE 7-22 Risk Category II—anticipated for this coastal location. Verify site terrain and obstruction characteristics on final survey.

Wind Wind design at 92 mph (3-second gust) anticipated—standard ASCE 7-22 wind load procedures recommended. No hurricane or extreme wind zone exposure anticipated for this location.

Snow Ground snow load of 5.8 psf anticipated—modest snow loading. Standard roof design methodology likely adequate; consult engineer on final roof slope and drainage.

INVESTIGATIONS & INSPECTIONS

Geotech Required Yes—geotechnical investigation recommended to confirm Site Class D assumption, obtain seismic parameters (Ss, S1, SDC), and establish design bearing capacity and settlement estimates.

Liquefaction Screen Liquefaction screening recommended—Site Class D + coastal location warrant evaluation. Groundwater depth and soil composition should be confirmed during geotechnical investigation.

Special Inspections Special inspections anticipated if SDC C or higher is confirmed; not anticipated under SDC B. Confirm with structural engineer once seismic category is established.

Code Edition ASCE 7-22 / IBC 2021 (planning reference only)

Site Class Note Site Class D assumed for planning purposes based on soil type and drainage characteristics. Geotechnical investigation recommended to confirm final site class and obtain seismic design parameters required for code compliance.

SOILS — USDA SOIL SURVEY

MAP UNIT: Coombs gravelly loam, 2 to 5 percent slopes **SERIES:** Coombs

DRAINAGE CLASS: Well drained **HYDRO GROUP:** C

SOIL ORDER: Alfisols **SUBORDER:** Xeralfs

AI SOIL INTELLIGENCE — CLAUDE HAIKU

The Coombs gravelly loam is a well-drained Alfisol formed in alluvial and colluvial deposits on gentle slopes in the Coast Range and inland valleys of California. This soil series features a gravelly loam surface layer overlying a clay-enriched substratum typical of Xeralfs, which develop under Mediterranean climates with dry summers. Surveyors and developers should expect moderate gravel content affecting excavation and compaction, good drainage reducing foundation concerns, but potential seasonal moisture retention in the clay substratum that may influence fill placement and long-term stability.

Gravelly alluvial and colluvial parent material on 2–5% footslope and terrace positions; typically supporting oak woodland and grassland vegetation.

DRAINAGE

Class Well drained

Seasonal Saturation None within 60 inches

Flooding Risk Minimal on 2–5% slopes; assess site-specific drainage patterns

Ponding Risk None

ENGINEERING

Bearing Capacity 2,500–3,500 psf (shallow foundations, gravel-rich upper soil)

Rating Good

Shrink-Swell Low to Moderate — clay substratum present but not dominant

Frost Action Low

Cut Suitability Good — gravelly material is stable and drains readily

Fill Suitability Fair — clay substratum requires moisture conditioning and proper compaction

SEPTIC

Suitability Conditional

Notes Perc test required — clay loam substratum may limit percolation rates; depth to limiting layer and seasonal saturation must be verified

HAZARDS

Excavation Moderate — 15–30% gravel content; equipment wear and slower production expected

Expansive Clay Low — clay content insufficient for significant expansion

Erosion Low on intact 2–5% slopes; moderate risk in disturbed areas and concentrated runoff paths

Data sources: USGS National Water Information System · [waterdata.usgs.gov](#) · California DWR / CDEC · [cdec.water.ca.gov](#) · USBR RISE · [data.usbr.gov](#) · USACE CWMS · [water.usace.army.mil](#) · USGS NWIS Groundwater · [waterservices.usgs.gov](#) · USDA Web Soil Survey · [sdmdataaccess.nrcs.usda.gov](#) · CA DWR SGMA Bulletin 118 · [gis.water.ca.gov](#) · ASCE 7-22 Hazard Tool · [gis.asce.org](#) · USGS Design Maps · [earthquake.usgs.gov](#) · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision

Water levels current as of report run: Jul 12, 2026, 9:43 PM UTC

* Live pool elevation not reported by this reservoir — full pool design elevation shown

 OIL & GAS WELLS

California — within 10mi radius

8

total

0

active

7

plugged

By status:

7

Plugged

1

Idle

By type: 8 DH

Lease / Well	Status	Type	Operator	Depth	Dist
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
McKenzie-Hazard Syndicate	Plugged	DH	McKenzie-Hazard Syndicate	—	7.4 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.6 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.8 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	9.5 mi
Capell Oil Corp.	Idle	DH	Capell Oil Corp.	—	9.7 mi
Cordelia Oil & Gas Co.	Plugged	DH	Cordelia Oil & Gas Co.	—	12 mi

AI Assessment — Well Proximity

Well proximity is not a planning concern for this site. The nearest well is a plugged Mobil Oil Corporation dry hole located 7.2 miles away, well beyond any actionable review distance. No active producers, injection wells, or other well types exist within distances requiring environmental or surface rights assessment.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

200 **10** **108**
total in search area within 5 mi within 25 mi

Site Name	Status	Commodities	Distance
Roderick Sand and Gravel Pit	Occurrence	Sand/Gravel	1.6 mi
Basalt Rock Co	Producer	Clay	2 mi
Cicero Pumice Quarry	Occurrence	PUM	2.9 mi
Mount George King Pumice Deposit.	Occurrence	PUM	2.9 mi
Basalt Rock Co.	Occurrence	DIT	3.1 mi
Mt George Pumice Deposit	Producer	PUM	3.1 mi
Wilson Pumice Quarry	Occurrence	PUM	3.7 mi
Unknown	Occurrence	Mercury	4.4 mi
Silva Pumice Quarry	Occurrence	PUM	4.7 mi
Silva Pumice Quarry	Producer	PUM	5 mi
Walker Pumice Quarry	Occurrence	PUM	5.1 mi
Walker Pumice Quarry	Producer	PUM	5.1 mi
Mountain	Occurrence	Mercury	5.4 mi
Unknown	Occurrence	Mercury	7.6 mi
Unknown	Occurrence	Stone/Crushed	7.8 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Prospect	Occurrence	Chromium	8.8 mi
Unnamed Location	Occurrence	Chromium	9 mi
Unknown	Occurrence	Chromium	9.1 mi
Bella Oak	Past Producer	Mercury	9.4 mi
Unknown	Occurrence	MG	9.7 mi
Serres Pit	Past Producer	Sand/Gravel	10.4 mi
Unnamed Location	Unknown	Sand/Gravel	10.5 mi
Conn Valley	Occurrence	Manganese	10.6 mi

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

Site Planning Review – Mineral Resources Three active mineral producers operate within 5 miles of the project site. Basalt Rock Co (2 mi) produces clay and operates an occurrence for diatomite at 3.1 mi, while Silva Pumice Quarry (5 mi) and Mt George Pumice Deposit (3.1 mi) are active pumice producers. Plan for heavy truck traffic on local access roads during operating hours, potential dust generation from aggregate and clay operations, and noise from crushing or processing activities, particularly from the closer facilities.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

6 **6** **12**
Endangered Threatened Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	SCIENTIFIC NAME	STATUS
California tiger Salamander	<i>Ambystoma californiense</i>	Endangered
Conservancy fairy shrimp	<i>Branchinecta conservatio</i>	Endangered
Contra Costa goldfields	<i>Lasthenia conjugens</i>	Endangered
Soft bird's-beak	<i>Cordylanthus mollis</i> ssp. <i>mollis</i>	Endangered
Suisun thistle	<i>Cirsium hydrophilum</i> var. <i>hydrophilum</i>	Endangered
Vernal pool tadpole shrimp	<i>Lepidurus packardii</i>	Endangered
Alameda whipsnake (=striped racer)	<i>Masticophis lateralis euryxanthus</i>	Threatened
California red-legged frog	<i>Rana draytonii</i>	Threatened
Delta smelt	<i>Hypomesus transpacificus</i>	Threatened
Northern spotted owl	<i>Strix occidentalis caurina</i>	Threatened
Vernal pool fairy shrimp	<i>Branchinecta lynchi</i>	Threatened
western snowy plover	<i>Charadrius nivosus nivosus</i>	Threatened

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Implications Review – 38.3459, -122.3047 ****Section 7 Trigger & Most Planning-Relevant Species:**** Any federal action, funding, or permitting (including Army Corps Section 404 wetland permits, USFWS Section 10 take authorizations, or federal highway/infrastructure approvals) will trigger mandatory ESA Section 7 consultation. The most planning-relevant species for this inland Bay Area/Central Valley location are ****California tiger salamander**** and ****vernal pool species**** (fairy shrimp, tadpole shrimp), which indicate seasonal ponding or wetland habitat; ****California red-legged frog****, which requires perennial or semi-permanent water bodies; and ****Alameda whipsnake****, which prefers grassland-chaparral ecotones. ****Recommended Next Step:**** Conduct a Phase 1 environmental site assessment with wetland delineation and a habitat assessment by a qualified biologist to determine if critical habitat or suitable occupied habitat exists on-site; this will clarify whether formal Section 7 consultation, Habitat Conservation Plan (HCP) coverage, or incidental take permits are necessary before design or permitting proceeds. --- ***For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.***

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

57	5051	2026-07-04
Unique Species	Total Observations	Most Recent
COMMON NAME	SCIENTIFIC NAME	LAST OBS
American black nightshade	<i>Solanum americanum</i>	2026-05-29
Bearded Clover	<i>Trifolium barbigerum</i>	2026-05-15
bird's foot cliffbrake	<i>Pellaea mucronata</i>	2026-04-28
Blow Wives	<i>Achyrachaena mollis</i>	2026-05-20
blue oak	<i>Quercus douglasii</i>	2026-05-09
Bolander's Phacelia	<i>Phacelia bolanderi</i>	2026-04-27
box elder	<i>Acer negundo</i>	2026-05-18
Braun's Giant Horsetail	<i>Equisetum telmateia braunii</i>	2026-06-22
California bay	<i>Umbellularia californica</i>	2026-06-24
California beeplant	<i>Scrophularia californica</i>	2026-05-20
California black oak	<i>Quercus kelloggii</i>	2026-05-15
California buckeye	<i>Aesculus californica</i>	2026-06-02
California Dutchman's Pipe	<i>Aristolochia californica</i>	2026-05-26
California milkwort	<i>Rhinotropis californica</i>	2026-05-15
California mugwort	<i>Artemisia douglasiana</i>	2026-06-02
California poppy	<i>Eschscholzia californica</i>	2026-06-24
California Wild Rose	<i>Rosa californica</i>	2026-05-09
Canyon liveforever	<i>Dudleya cymosa</i>	2026-05-30
-	<i>Castilleja densiflora densiflora</i>	2026-05-09
clay mariposa lily	<i>Calochortus argillosus</i>	2026-05-15
coast live oak	<i>Quercus agrifolia</i>	2026-05-13
coastal woodfern	<i>Dryopteris arguta</i>	2026-05-20
common selfheal	<i>Prunella vulgaris</i>	2026-06-22
Diogenes' lantern	<i>Calochortus amabilis</i>	2026-05-15
distant phacelia	<i>Phacelia distans</i>	2026-05-15
-	<i>Downingia concolor concolor</i>	2026-05-15
Dwarf Brodiaea	<i>Brodiaea terrestris</i>	2026-05-15
Elegant Clarkia	<i>Clarkia unguiculata</i>	2026-05-16
goldback fern	<i>Pentagramma triangularis</i>	2026-05-20

COMMON NAME	SCIENTIFIC NAME	LAST OBS
Ithuriel's Spear	<i>Triteleia laxa</i>	2026-06-15
maroonspot calicoflower	<i>Downingia concolor</i>	2026-05-15
narrowleaf milkweed	<i>Asclepias fascicularis</i>	2026-06-25
narrowleaf mule-ears	<i>Wyethia angustifolia</i>	2026-05-15
Narrowleaf Willow	<i>Salix exigua</i>	2026-06-24
needleleaf navarretia	<i>Navarretia intertexta</i>	2026-05-15
northern California black walnut	<i>Juglans hindsii</i>	2026-06-24
orange bush monkeyflower	<i>Diplacus aurantiacus</i>	2026-05-29
Oregon Ash	<i>Fraxinus latifolia</i>	2026-06-02
Pacific Bleeding Heart	<i>Dicentra formosa</i>	2026-05-04
Pacific madrone	<i>Arbutus menziesii</i>	2026-07-04
Pacific poison oak	<i>Toxicodendron diversilobum</i>	2026-06-02
pineapple-weed	<i>Matricaria discoidea</i>	2026-05-20
Redwood Sorrel	<i>Oxalis smalliana</i>	2026-05-28
small-headed clover	<i>Trifolium microcephalum</i>	2026-05-15
Superb Mariposa Lily	<i>Calochortus superbus</i>	2026-06-11
thimble clover	<i>Trifolium microdon</i>	2026-05-15
tomcat clover	<i>Trifolium willdenovii</i>	2026-05-15
Toyon	<i>Heteromeles arbutifolia</i>	2026-06-19
trailing blackberry	<i>Rubus ursinus</i>	2026-06-24
Tree Clover	<i>Trifolium ciliolatum</i>	2026-05-15
valley oak	<i>Quercus lobata</i>	2026-06-24
wavy-leafed soap plant	<i>Chlorogalum pomeridianum</i>	2026-05-20
western sycamore	<i>Platanus racemosa</i>	2026-05-17
white brodiaea	<i>Triteleia hyacinthina</i>	2026-05-15
white-tipped clover	<i>Trifolium variegatum</i>	2026-05-15
Winecup Clarkia	<i>Clarkia purpurea</i>	2026-05-15
Yellow Mariposa Lily	<i>Calochortus luteus</i>	2026-06-11

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment: 38.3459, -122.3047 This site supports a ****mixed oak woodland and chaparral community****, characteristic of the North Bay foothills, with blue oak and California black oak as dominant canopy species alongside California bay and California buckeye in riparian or mesic microsites. The presence of moisture-dependent species like Braun's Giant Horsetail and clay mariposa lily indicates ****seasonal water availability with variable drainage****, suggesting the site includes both well-drained upland slopes and areas with spring/seepage flows. For grading and drainage planning, expect ****moderate to low permeability on clay soils (evidenced by mariposa lily and native clover presence) with seasonal water table fluctuations****—drainage design should preserve existing swales and account for wet-season saturation while avoiding hydrological disruption to the horsetail and riparian communities. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

635	154	91	1439
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants
VENOMOUS SNAKES			
SPECIES		OBSERVATIONS	
<i>Crotalus oreganus oreganus</i>		635 obs	
VENOMOUS SPIDERS			
SPECIES		OBSERVATIONS	
<i>Latrodectus hesperus</i>		154 obs	
STINGING INSECTS			
SPECIES		OBSERVATIONS	
<i>Dasymutilla aureola</i>		91 obs	
<i>Dasymutilla californica</i>		91 obs	
<i>Dasymutilla sackenii</i>		91 obs	
TOXIC PLANTS			
SPECIES		OBSERVATIONS	
<i>Toxicodendron diversilobum</i>		1100 obs	

SPECIES	OBSERVATIONS
<i>Datura wrightii</i>	57 obs
<i>Datura stramonium</i>	57 obs
<i>Datura ferox</i>	57 obs
<i>Conium maculatum</i>	199 obs
<i>Urtica gracilis</i>	83 obs
<i>Urtica gracilis holosericea</i>	83 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Briefing – Northern California Survey Area ****Highest-Priority Hazards:**** Toxic plants dominate this region (1,439 observations), particularly poison oak, which causes severe skin reactions and is present year-round. Northern Pacific rattlesnakes are the second critical concern with 635 documented sightings; black widow spiders and stinging wasps pose moderate but manageable risks. ****Practical Avoidance:**** Wear long pants, closed-toe boots, and gloves; wash exposed skin immediately after fieldwork to prevent poison oak exposure. Stay alert for rattlesnakes in rocky, brushy areas—make noise while moving and watch where you place your hands. Check gear and clothing before leaving site, as spiders and wasps can travel with equipment. Avoid touching any plants you cannot confidently identify. ****Seasonal Considerations:**** Rattlesnake activity peaks spring through fall; poison oak remains hazardous year-round but causes worse reactions during active growth (spring/summer). Wasps are most aggressive in late summer and fall. Always verify current conditions locally before fieldwork.

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

3	545ft	3
Beam Paths	Closest Beam	Fixed P2P

TX CALL	RX CALL	TYPE	PATH MI	CLOSEST
WBO61	WBO59	Fixed Point-to-Point	—	545ft
KYH37	WQAT627	Fixed Point-to-Point	—	927ft
WRCR329	WQAT627	Fixed Point-to-Point	—	1289ft

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

FAA AIRWAYS — ATS Routes crossing site area

26	13	2	11
Total Airways	Jet Routes	Victor Airways	Other Routes

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	303.8°	121.4°
J110	Jet Route	High	18,000 ft	—	137.9°	318.2°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	18,000 ft	—	317.1°	135.4°
J58	Jet Route	High	18,000 ft	—	65.3°	245.9°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
J80	Jet Route	High	18,000 ft	—	65.3°	245.9°
J84	Jet Route	High	18,000 ft	—	50.8°	231.0°
J88	Jet Route	High	18,000 ft	—	308.0°	127.3°
J94	Jet Route	High	18,000 ft	—	65.3°	245.9°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	161.4°	341.7°
T257	T-Route	Low	—	—	321.1°	140.5°
T259	T-Route	Low	—	—	41.6°	221.8°
T261	T-Route	Low	—	—	31.4°	211.7°
T263	T-Route	Low	—	—	308.8°	128.2°
V107	Victor Airway	Low	7,000 ft	—	295.4°	114.2°
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°

RAILROAD PROXIMITY — FRA North American Rail Network

8
Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	Other	1	No	No	—
CFNR	Yard	1	No	No	—
CFNR	Mainline	1	No	No	VALLEJO
SMRT	Mainline	1	No	No	BRAZOS JCT.
NVRR	Mainline	1	No	No	MARTINEZ
USG	Other	1	No	No	—
CFNR	S	1	No	No	—
NVRR	Yard	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200 91 66 5 14 6
Total Observations Total Species Birds Mammals Reptiles Amphibians

BIRDS (66 species)

COMMON NAME	Scientific Name	OBS
Wild Turkey	<i>Meleagris gallopavo</i>	9
Snowy Egret	<i>Egretta thula</i>	7
California Quail	<i>Callipepla californica</i>	5
Western Bluebird	<i>Sialia mexicana</i>	5
California Towhee	<i>Melospiza crissalis</i>	5
Great Egret	<i>Ardea alba</i>	4
Red-tailed Hawk	<i>Buteo jamaicensis</i>	4
Black Phoebe	<i>Sayornis nigricans</i>	4
Western Screech-Owl	<i>Megascops kennicottii</i>	4
Cooper's Hawk	<i>Astur cooperii</i>	3
Anna's Hummingbird	<i>Calypte anna</i>	3
Ash-throated Flycatcher	<i>Myiarchus cinerascens</i>	3
Turkey Vulture	<i>Cathartes aura</i>	3
Mallard	<i>Anas platyrhynchos</i>	3
Bushtit	<i>Psaltiriparus minimus</i>	3

+ 51 additional species

MAMMALS (5 species)

COMMON NAME	Scientific Name	OBS
Black-tailed Jackrabbit	<i>Lepus californicus</i>	4
Coyote	<i>Canis latrans</i>	2
Mule Deer	<i>Odocoileus hemionus</i>	2
Eastern Fox Squirrel	<i>Sciurus niger</i>	1
Striped Skunk	<i>Mephitis mephitis</i>	1

REPTILES (14 species)

COMMON NAME	Scientific Name	OBS
Western Fence Lizard	<i>Sceloporus occidentalis</i>	9
Pacific Gopher Snake	<i>Pituophis catenifer catenifer</i>	8
Northern Pacific Rattlesnake	<i>Crotalus oreganus oreganus</i>	6
California King Snake	<i>Lampropeltis californiae</i>	3
Pond Slider	<i>Trachemys scripta</i>	2
Western Skink	<i>Plestiodon skiltonianus</i>	2
Ring-necked Snake	<i>Diadophis punctatus</i>	1
Common Garter Snake	<i>Thamnophis sirtalis</i>	1

COMMON NAME	Scientific Name	OBS
Western Yellow-bellied Racer	Coluber constrictor mormon	1
Pacific Ringneck Snake	Diadophis punctatus amabilis	1
Northwestern Pond Turtle	Actinemys marmorata	1
Common Sharp-tailed Snake	Contia tenuis	1
North American Racer	Coluber constrictor	1
Southern Alligator Lizard	Elgaria multicarinata	1

AMPHIBIANS (6 species)

COMMON NAME	Scientific Name	OBS
Pacific chorus frog	Pseudacris regilla	5
American Bullfrog	Lithobates catesbeianus	3
Foothill Yellow-legged Frog	Rana boylei	3
Rough-skinned Newt	Taricha granulosa	1
Western Toad	Anaxyrus boreas	1
California Giant Salamander	Dicamptodon ensatus	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment This site in Sonoma County, California exhibits characteristics of oak woodland and grassland habitat with proximal aquatic resources, as evidenced by a diverse assemblage including wetland-dependent species (egrets, pond turtles, chorus frogs) alongside upland generalists (quail, towhee, jackrabbit). The wildlife community includes several species of regulatory concern: Foothill Yellow-legged Frog (CDFW Species of Special Concern), Northwestern Pond Turtle (CDFW Species of Special Concern), and California Giant Salamander (CDFW Species of Special Concern), suggesting presence of high-quality riparian or seasonal wetland habitat warranting avoidance or careful evaluation. The abundance of native predators (hawks, coyote) and diverse reptile assemblage indicates relatively low-to-moderate disturbance levels with functional ecological structure, though the presence of Eastern Fox Squirrel and American Bullfrog suggests some prior landscape modification. This wildlife composition indicates a semi-natural mosaic likely supporting seasonal wetlands, perennial or intermittent streams, and oak savanna, requiring hydrological sensitivity and habitat connectivity considerations in any development planning. For planning purposes only. Consult a qualified biologist for site-specific biological assessments.

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

✳️ PASSIVE + ACTIVE SOLAR PLANNING — Lat 38.35°

1 · AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 · IDEAL SOLAR ARRAY PITCH

Optimal tilt = 38.3° (latitude) → 9/12 pitch

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	97%	good
5/12	22.6°	97.6%	good
6/12	26.6°	98.2%	↔ excellent
7/12	30.3°	98.8%	↔ excellent
8/12	33.7°	99.3%	↔ excellent
9/12	36.9°	99.8%	↔ annual optimal
10/12	39.8°	99.8%	↔ excellent
12/12	45°	99%	↔ excellent

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~\$108 value/yr). Over 25yr panel life: ~\$2710 total gain.

3 · SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	51	31	Minimize south glass
South glass (SHGC 0.6)	76	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	27	104	⚠ Minimize — afternoon load
Roof — dark (absorb 0.95)	236	301	Highest load surface
Roof — light (absorb 0.35)	87	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	62	79	Best roof option for heat
South wall — dark	94	69	
South wall — light	31	23	Light color = major savings
Hardscape — asphalt	181	301	⚠ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr·ft² additional load per 100sf of adjacent hardscape.
Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 · ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 • THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 • TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.1	1.9	2.4	OK
30ft	All year	3.1	2.9	3.7	Zone 2 check
40ft	All year	4	3.8	5.2	△ verify
50ft	All year	4.8	4.6	7.3	△ verify
60ft	All year	5.6	5.5	9.3	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.

Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.

△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 • ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45

Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45

Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor

†**Balanced** = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)

WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.

△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

🏠 AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar & Energy Plan You're in a great location for solar—you get about 6.6 hours of useful sunlight per day on average, which means your panels will produce solid energy year-round, even though winter dips down to just under 4 hours. Your roof ideally tilts at about 38 degrees (roughly a 9/12 pitch), and here's where roof color becomes surprisingly important: a dark roof soaks up over 300 BTUs of heat per hour per square foot in summer, while a light-colored roof takes in less than half that, which can save you real money on cooling. West-facing windows are your biggest summer problem—they pour in over 100 BTUs of afternoon heat, so you'll want to shade those with overhangs, blinds, or trees if possible. A big deciduous tree on your south side is actually your friend: it blocks summer sun when you don't want it (about 4 hours on a typical day) but drops its leaves in winter to let warmth in when you need it most. Thermal mass is simply heavy materials like concrete or masonry that slowly absorb heat during the day and release it at night, smoothing out temperature swings. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.



Technical Brief — for architects, engineers, and surveyors

SITE INTELLIGENCE BRIEFING **38.34586°N, 122.30466°W | Napa County, California**

Structural Design Drivers & Climate Envelope

This site falls within IECC Zone 3C (Marine), characterized by an Annual Freezing Index of 0°F-days, indicating no meaningful frost cycling and negligible freeze-thaw design burden on exposed surfaces or shallow utilities. Wind, snow, and seismic loading parameters are not available in the provided dataset; however, the marine climate designation and zero AFI suggest this coastal-influenced location will require wind speed and snow load verification through ASCE 7 or current California Building Code provisions before structural system selection proceeds. The absence of reported snow and wind data does not eliminate these loads—it flags them as the first required research items before member sizing or lateral system design.

Geotechnical Foundation Design

Subsurface conditions are mapped as Coombs gravelly loam, 2 to 5 percent slopes, classified as Hydrologic Group C with well-drained drainage characteristics. Group C soils typically support adequate bearing capacity and present no acute consolidation or settlement risk when properly evaluated; however, the gravelly loam texture and well-drained classification indicate shallow groundwater infiltration potential and possible seasonal perched water tables in low-lying areas. Frost depth is not reported, consistent with the 0°F-day AFI—footings need not extend below frost line for frost protection, but structural requirements and bearing capacity will govern minimum depth. A site-specific geotechnical investigation including SPT or CPT, moisture content sampling, and bearing capacity analysis is standard for any structure; the Coombs series does not preclude conventional spread footings or grade beams, but permeability and slope drainage must be verified in design.

Hazard Assessment

No Quaternary faults are mapped within the search radius, and no subsidence basin data has been identified, eliminating fault rupture and regional subsidence as primary design drivers for this location. Wildfire risk data was not provided in the briefing dataset. Seismic hazard parameters are not included; California seismic design requirements (California Building Code Chapter 11, per ASCE 41 and ground motion maps) must be applied independently and are not cleared by the absence of local fault mapping.

Site Context & Mining Proximity

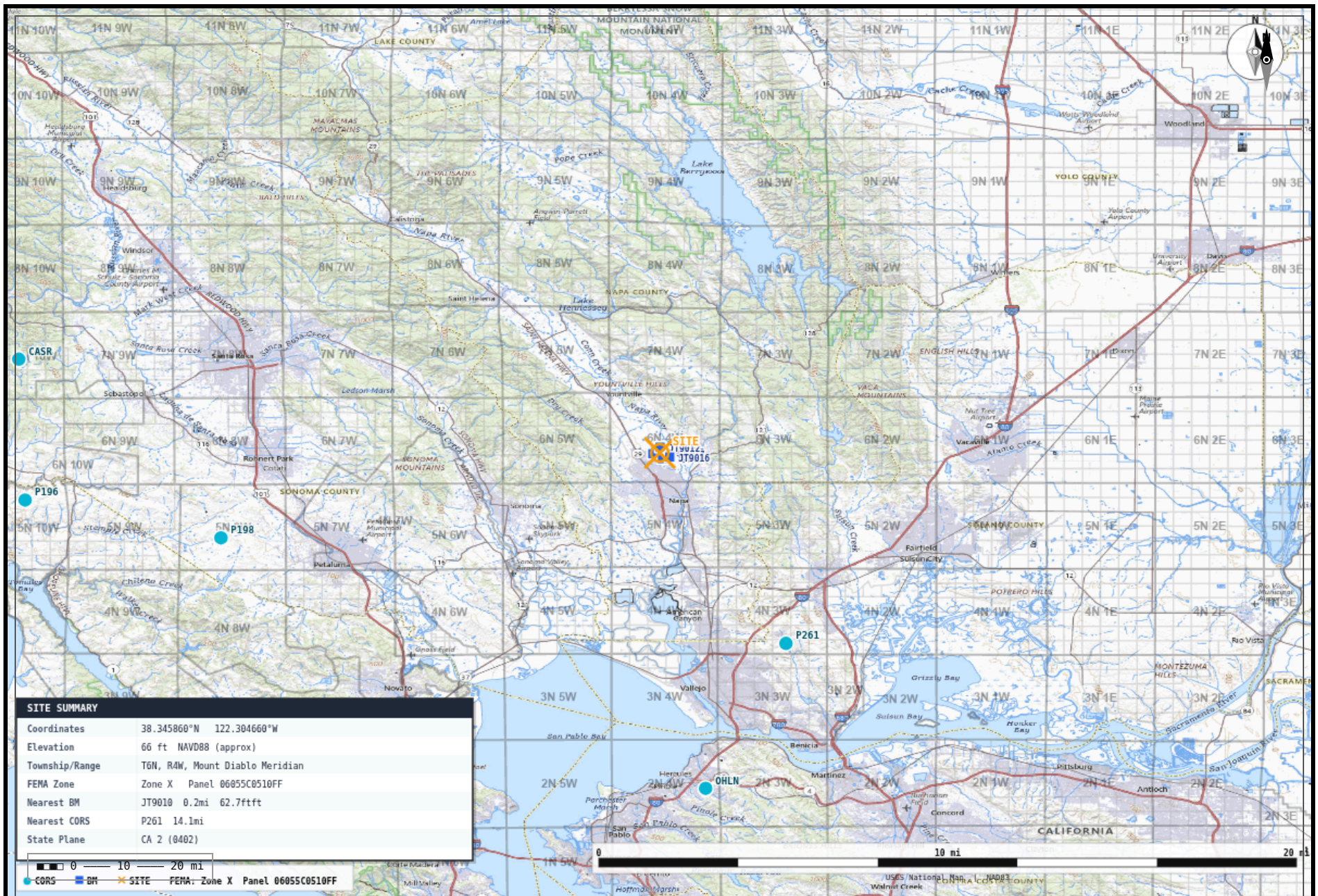
The site is situated within a historically and actively mined region: three active mineral producers operate within 5 miles (Basalt Rock Co. at 2 miles producing clay; Mt George Pumice Deposit at 3.1 miles; Silva Pumice Quarry at 5 miles). While these operations are outside the immediate development footprint, vibration transmission from blasting, dust control requirements, and long-term land stability in the broader area warrant review of mining operation permits and historical activity records. Approximately 200 MRDS deposit locations exist within ~35 miles, confirming a mature extraction landscape. No active oil or gas producers are present within the 10-mile search radius; the nearest well (Mobil Oil Corporation, 7.2 miles) is plugged and not a current pressure or contamination vector.

****Bottom Line:**** Obtain current seismic design parameters (response spectra, site class) and wind/snow loads per ASCE 7 and CBC before advancing structural design; perform site-specific geotechnical investigation to confirm Coombs loam properties, bearing capacity, and slope stability on the 2–5% grade.



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Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.

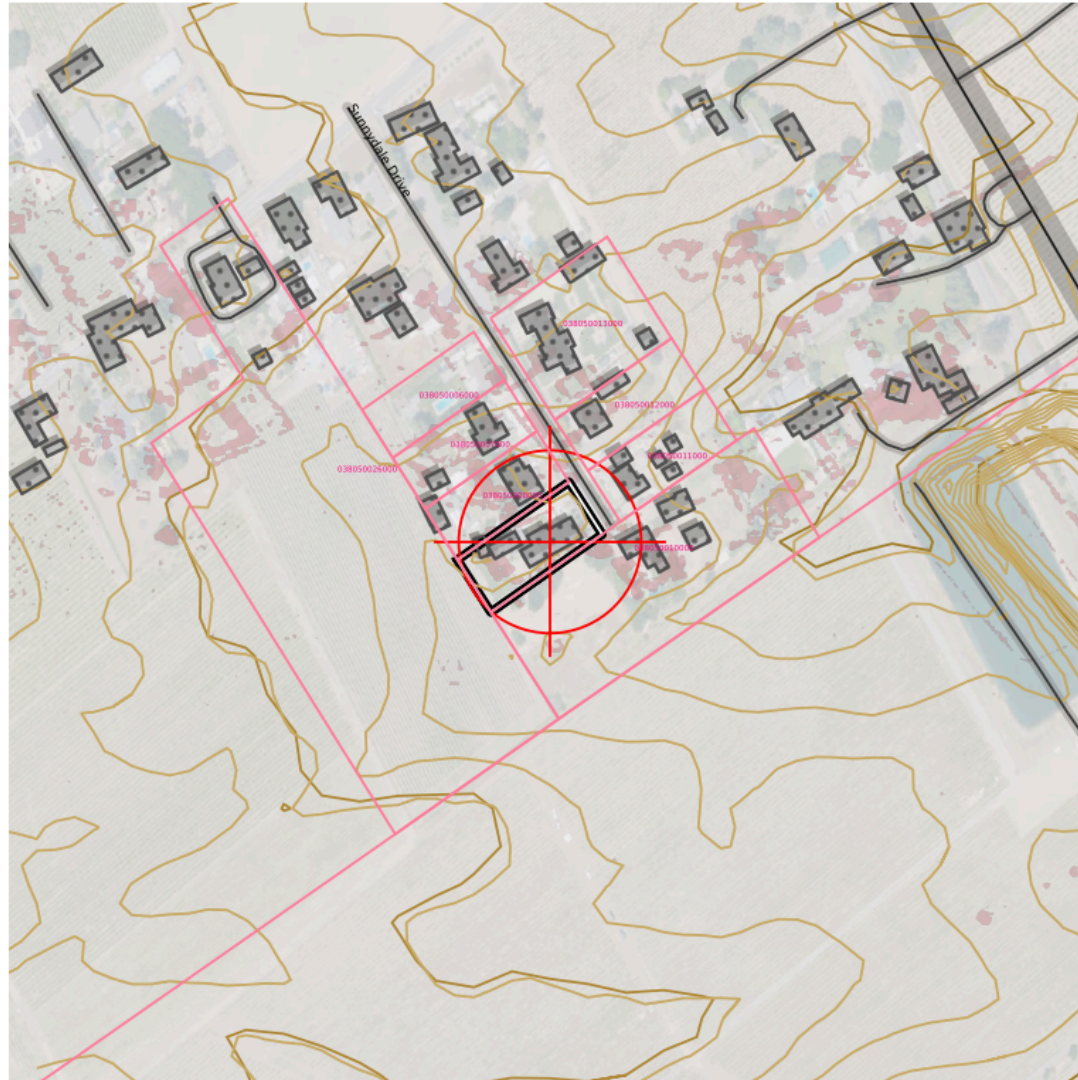


TOPOSCOUT
FIELD MAP

TopoScout Field Map
Lat: 38.34586 Lng: -122.30466
38.345860°N 122.304660°W | Radius: 30 mi

Generated: July 12, 2026
Base: USGS National Map
Generated by TopoScout - outkastdesigns.com

TOPOSCOUT SITE PLAN PREVIEW | \$19 UPGRADE FOR AUTOCAD DXF FILE



INCLUDED LAYERS



0 — 400 ft State Plane · NAD83 · US Feet

- Contours (Index/Major/Minor/1ft)
- PLSS Section/Township/QQ
- Roads Major/Local/Trail + ROW
- Buildings
- Hydro
- Power Lines
- Pipeline
- Railroad
- Benchmarks + Labels
- County Boundary
- OTLS Grid
- Site Info
- Elevation Labels
- 3D Polylines (all geometry)

✂ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

POST-PROCESSING

NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

CORS Data Download – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Raw RINEX observation files from CORS stations

FEMA & FLOOD

FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – [https://www.fema.gov/flood-](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

[insurance/work-with-nfip/elevation-certificate](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

Current EC form, instructions, LOMA/LOMR submission guidance

BASE STATION & NETWORK

NOAA CORS Network – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Continuously operating reference stations, data download

NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

REFERENCE & CALCULATIONS

NGS Datasheets – [https://www.ngs.noaa.gov/cgi-](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

[bin/datasheet.prl](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

NGS Geoid Models – <https://geodesy.noaa.gov/GEOID/>

GEOID18 and current model downloads, GEOID calculator

DRONE OPERATIONS

FAA Part 107 — Commercial Rules –

https://www.faa.gov/uas/commercial_operators

Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers –

https://www.faa.gov/uas/commercial_operators/part_107_waivers

Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

[AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

[AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

[NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

